

ACADEMIC BLOCKCHAIN & AGENTIC CONSORTIUM

Engineering Task Allocation & Work Breakdown Plan — 6 Groups (18 Contributors)

3-Organization Hyperledger Fabric 2.5 Consortium • IPFS Storage • Multi-Agent Automation Ecosystem

Project Baseline & Workload Distribution Principle:

All groups inherit the operational baseline (3-Org Fabric 2.5 network, IPFS cluster, base chaincode, REST API & Portal) and build outward. To maintain strict equality, every group is assigned a dedicated Blockchain deliverable alongside their specialized Agentic Module.

Group 1: Academic Department Ecosystem & Decentralized Identity

Org1MSP • Academic Dept • Stages 1-3

Contributor 1.1 — Academic Chaincode Lead

- **Blockchain:** Enhances RegisterStudent and RequestCertificate chaincode functions with rich schema validation.
- **Module/Agent:** Connects student registration and request submission API routes with prerequisite validation rules.

Contributor 1.2 — CouchDB State & Identity Lead

- **Blockchain:** Designs CouchDB0 rich query indexes for department data, manages Org1MSP cryptographic identities & TLS.
- **Module/Agent:** Implements multi-level departmental approval pipeline (Faculty, HOD, DAC) in business layer.

Contributor 1.3 — Org 1 Academic Agent Lead

- **Blockchain:** Integrates Fabric Gateway Node SDK to submit signed Org1 transactions to peer0.org1:7051.
- **Module/Agent:** Builds Org 1 Autonomous Department Agent auto-verifying credits, attendance & prerequisites for stages 1–3.

Group 2: Autonomous Multi-Agent Inter-Organizational Verification & Consensus

Org Endorsement • Agent Handshakes • Central Engine

Contributor 2.1 — Inter-Org Consensus Chaincode Lead

- **Blockchain:** Implements state machine chaincode & enforces multi-org cryptographic endorsement policies on Fabric.
- **Module/Agent:** Builds multi-organization transaction coordinator handling stage progression across Org1, Org2, Org3.

Contributor 2.2 — Cryptographic Handshake Protocol Engineer

- **Blockchain:** Implements cross-peer endorsement gathering & cryptographic signature validation from all 3 peer nodes.
- **Module/Agent:** Develops Inter-Organizational Handshake Protocol for autonomous agents to exchange signed verification tokens.

Contributor 2.3 — Master Multi-Agent Orchestration Architect

- **Blockchain:** Anchors agent execution audit trails, timestamps, and consensus proofs directly onto the Fabric ledger.
- **Module/Agent:** Builds Master Multi-Agent Orchestrator supporting 1-Click Autonomous Execution & reasoning audit logs.

Group 3: Intelligent Fraud Detection, Anomaly Analysis & Security Engine

On-Chain Invariants • Fraud Detection • Risk Engine

Contributor 3.1 — Ledger Invariant & State Auditor

- **Blockchain:** Develops chaincode audit functions to inspect historical ledger state for invariant violations and tampering.
- **Module/Agent:** Connects backend fraud inspection triggers with live Fabric ledger query mechanisms.

Contributor 3.2 — Transcript Anomaly & Heuristics Lead

- **Blockchain:** Analyzes on-chain student transcript records to detect statistical grade inflation and CGPA spikes.
- **Module/Agent:** Develops velocity anomaly rules (detecting request replay attacks, rapid submissions, collision checks).

Contributor 3.3 — Autonomous Fraud Scoring Lead

- **Blockchain:** Integrates smart contract quarantine flags to lock suspicious certificate requests on the Fabric ledger.
- **Module/Agent:** Builds Autonomous Fraud Detection Agent and unified 0–100 Risk Engine (LOW, MEDIUM, HIGH, CRITICAL).

Group 4: Cryptographic Ledger Proofs, IPFS Storage & Verifiable QR Engine

Ledger Proofs • IPFS Kubo • QR Security

Contributor 4.1 — Ledger Hash Anchoring Engineer

• **Blockchain:** Implements chaincode functions to anchor IPFS CIDs and SHA-256 digests onto immutable Fabric ledger.

• **Module/Agent:** Integrates IPFS Kubo node API (5001/8080) for decentralized PDF storage, automatic pinning, and retrieval.

Contributor 4.2 — Verifiable QR & PDF Cryptography Lead

• **Blockchain:** Encodes tamper-proof cryptographic metadata (Cert ID, IPFS CID, SHA-256, TxID) into QR payload.

• **Module/Agent:** Generates official academic certificates with embedded verifiable QR codes using pdfkit & qrcode.

Contributor 4.3 — QR Scanner & Ledger Decryptor

• **Blockchain:** Implements backend QR verification endpoint (/api/verify/qr) cross-verifying QR against Fabric ledger.

• **Module/Agent:** Builds browser-based live Camera & File QR Scanner supporting 1-click instant verification.

Group 5: Ledger-Aware Conversational AI Assistant & Blockchain Event Hub

Fabric Events • History Query • Chatbot Agent

Contributor 5.1 — Fabric Event & History Hub Specialist

• **Blockchain:** Implements Fabric Gateway gRPC event listeners (Block & Commit Listeners) and GetHistoryForKey query.

• **Module/Agent:** Serializes raw blockchain history and transaction payloads into structured data for agent consumption.

Contributor 5.2 — Conversational NLP & Intent Engine Lead

• **Blockchain:** Maps natural language queries (e.g. "Check status of CERT-101") into blockchain query parameters.

• **Module/Agent:** Implements Conversational AI Agent (Chatbot Engine) providing intelligent natural language explanations.

Contributor 5.3 — Chatbot Widget & Real-Time Client Lead

• **Blockchain:** Integrates real-time ledger updates into the chat context so the chatbot reflects live blockchain state.

• **Module/Agent:** Builds responsive floating/docked Chatbot UI Widget with quick-action prompt chips & history.

Group 6: Consortium Consensus, Health Monitoring & DevOps Orchestration

Raft Consensus • Health Monitor • Master QA

Contributor 6.1 — Raft Consensus & Network Topology Engineer

• **Blockchain:** Manages Raft ordering service (7050), channel creation, and configtx.yaml consortium definitions.

• **Module/Agent:** Ensures zero-downtime cross-platform Docker containerization across macOS and Ubuntu Linux.

Contributor 6.2 — Autonomous Health Monitoring Agent Lead

• **Blockchain:** Probes peer gRPC ports, orderer status, CouchDBs, IPFS, and cross-peer block height sync parity.

• **Module/Agent:** Builds System Health Monitoring Agent with real-time telemetry dashboard & self-healing advice.

Contributor 6.3 — Full-Stack API Gateway & Master QA Lead

• **Blockchain:** Configures unified Fabric gateway connection pooling across all 3 orgs for Express backend.

• **Module/Agent:** Builds master web portal navigation & automated integration test runner (test-flow.sh).

EQUAL WORKLOAD & DELIVERABLES MATRIX

Group	Focus Area	Blockchain Deliverable	Agentic / Module Deliverable
Group 1	Academic Org & Identity	Stages 1-3 Chaincode & CouchDB0 Indexes	Org 1 Autonomous Academic Agent
Group 2	Autonomous Agent Engine	Multi-Org Endorsement Chaincode & Handshake	Master Autonomous Multi-Agent Orchestrator
Group 3	Fraud Detection & Anomaly	On-Chain Audit Invariants & State Hash Checks	Autonomous Fraud Detection & 0-100 Risk Engine
Group 4	QR & Storage	SHA-256 / CID Ledger Anchoring & Proofs	Verifiable QR PDF Generator & Scanner
Group 5	Ledger NLP AI	gRPC Event Subscriptions & GetHistoryForKey	Conversational AI Chatbot & Intent Engine
Group 6	DevOps & Health	Raft Consensus, Channel Topology & Sync	Health Monitoring Agent & Master QA

PROJECT PHASES & IMPLEMENTATION MILESTONES

1. Phase 1: Baseline Network Validation: Verify existing 3-Org Fabric network, cryptogen, channel participation, and IPFS node.
2. Phase 2: Modular Expansion & Development: Groups 1-6 concurrently develop their Blockchain methods and Agentic Modules.
3. Phase 3: Master Integration & Control Center: Bind all agent endpoints, QR scanner, Health telemetry, and Chatbot to master UI.
4. Phase 4: Consortium Verification & Evaluation: Execute master test suite (test-flow.sh) validating multi-agent handshakes & tamper-proofing.